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COP 3035C-042
Homework 3

Problem

The task is to build a study tracker where the user can enter the amount of time they spent studying doing various activities and receive an assessment of their engagement level. I have to choose activities and weights for each activity. I need to prompt the user to get the times for individual sessions for each activity. I need to get the average time spent per session for each activity. I need to get a weighted score by multiplying the average by the weight for each activity. I need to sum up all the weighted scores and use that to get an engagement level score. The engagement level score is high if the sum of the weighted scores is greater or equal to 80, medium if it is less than 80 and greater than or equal to 60, and low if it is less than 60. I need to display all this information in an organized report.

Solution

I chose the categories: reading, coding, video, and quiz. I chose the weights: 0.25, 0.35, 0.2, and 0.2 respectively. I used a function called `get_category_data` that takes a `category_name` as input to get input from the user. The user inputs the amount of time spent per session for the given category. The function returns three values: `session_count`, `total_minutes`, and `values_str`. `session_count` is the number of sessions the user gave, `total_minutes` is the sum of all the session times, and `values_str` is a string that has the session times separated by commas to be used for display.

There is a function, `weighted_contribution` that takes the average time among a given category's session times and the weight for the given category and multiplies those together and returns the value. That is used as the weighted score. There is also an `engagement_level` function. It takes in `total_score` which is all the weighted scores for all activities summed up and it returns either "High", "Medium", or "Low" as described in the Problem section.

In the main function the categories and weights are defined. The `total_score` is initialized to 0 and I initialize two empty strings called `minutes_section` and `weight_section`. Those will be used to combine text from each category to be used to display the report. Then I loop through the categories by using the range function and the number of categories. I collect the necessary data using the functions mentioned above, compute the average and update the `total_score`. Then I format the data into strings to be added to the `minutes_section` and `weight_section` strings. After the loop I get the engagement score, create the strings for the last section of the report and print the entire report.

Code

```
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# COP3035-042
# Homework 3

def main():
    study_categories = ('reading', 'coding', 'video', 'quiz')
    category_weights = (0.25, 0.35, 0.2, 0.2)

    total_score = 0
    minutes_section = ''
    weight_section = ''
    for i in range(len(study_categories)):
        category = study_categories[i]
        weight = category_weights[i]

        # get data from user and compute average and weighted score
        session_count, total_minutes, values_str =
get_category_data(category)
        if session_count == 0:
            continue
        average = total_minutes / session_count
        weighted_score = weighted_contribution(average, weight)
        total_score += weighted_score

        # construct and store text for first part of report
        category_text = category.capitalize()
        minute_calculation_text = f'{total_minutes} / {session_count}'
        minutes_section += f'{category_text} + "
minutes":<22}{values_str:<18}= {minute_calculation_text:>7}
={average:>6.2f}\n'

        # construct and store text for second part of report
        weight_calculation_text = f'{average:.2f} x {weight:.2f}'
        weight_section += f'{category_text:<22}:
{weight_calculation_text:>12} = {weighted_score:>5.2f}\n'

    # assemble report
    engagement = engagement_level(total_score)
```

```

    total_report_section = f'{"TOTAL WEIGHTED SCORE":<22}:'
{total_score:<46.2f}\n'
    total_report_section += f'{"ENGAGEMENT LEVEL":<22}: {engagement:<46}'

# output report
print()
print('STUDY TRACKER REPORT')
print('-' * 70)
print(minutes_section[:-1])
print('-' * 70)
print(weight_section[:-1])
print('-' * 70)
print(total_report_section)
print('-' * 70)

def get_category_data(category_name):
    input_count = 1
    values_str = ''
    total_minutes = 0
    session_count = 0

    print(f'Enter {category_name} session lengths in minutes. Enter "x" to
finish category.')
    while(True):
        session_length = input(f'How long was {category_name} session
{input_count} in minutes? ')

        if session_length == 'x':
            values_str = values_str[:-2]
            break
        elif session_length.isdigit():
            values_str = values_str + session_length + ', '
            total_minutes = total_minutes + int(session_length)
            session_count = session_count + 1
            input_count += 1
        else:
            print('Invalid input')
            continue
    return session_count, total_minutes, values_str

```

```
def weighted_contribution(avg, weight):  
    return avg * weight  
  
def engagement_level(total_score):  
    if total_score >= 80:  
        return 'High'  
    elif total_score >= 60:  
        return 'Medium'  
    else:  
        return 'Low'  
  
if __name__ == "__main__":  
    main()
```

Output

```
Enter reading session lengths in minutes. Enter "x" to finish category.
How long was reading session 1 in minutes? 30
How long was reading session 2 in minutes? 45
How long was reading session 3 in minutes? 60
How long was reading session 4 in minutes? x
Enter coding session lengths in minutes. Enter "x" to finish category.
How long was coding session 1 in minutes? 60
How long was coding session 2 in minutes? 75
How long was coding session 3 in minutes? x
Enter video session lengths in minutes. Enter "x" to finish category.
How long was video session 1 in minutes? 20
How long was video session 2 in minutes? 0
How long was video session 3 in minutes? 15
How long was video session 4 in minutes? x
Enter quiz session lengths in minutes. Enter "x" to finish category.
How long was quiz session 1 in minutes? 10
How long was quiz session 2 in minutes? 10
How long was quiz session 3 in minutes? 5
How long was quiz session 4 in minutes? x

STUDY TRACKER REPORT
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Reading minutes      30, 45, 60      = 135 / 3 = 45.00
Coding minutes       60, 75          = 135 / 2 = 67.50
Video minutes        20, 0, 15       = 35 / 3 = 11.67
Quiz minutes         10, 10, 5       = 25 / 3 = 8.33
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Reading              : 45.00 x 0.25 = 11.25
Coding               : 67.50 x 0.35 = 23.62
Video                : 11.67 x 0.20 = 2.33
Quiz                 : 8.33 x 0.20 = 1.67
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TOTAL WEIGHTED SCORE : 38.88
ENGAGEMENT LEVEL    : Low
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```